EECE 571Q

Lecture 09: Beneath the surface (part 1)

Tuesday 9 June 2026

Announcements

- Quiz 4 today
- Project proposal due Friday 12 June 23:59 (information available on PrairieLearn)
- Final exam reference sheet info distributed later this week (must be submitted for review at least 3 days before exam)

Last time

We discussed various approaches to performing state preparation and measurement fault-tolerantly, and the challenges involved.

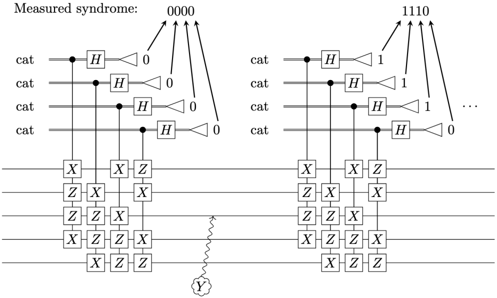


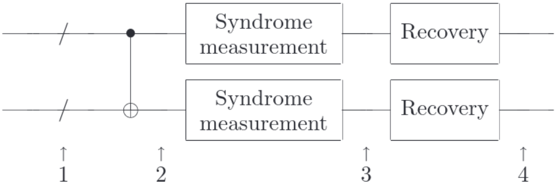
Figure 12.7: Measuring the full syndrome and then repeating for the five-qubit code. When the Y_3 error occurs after the first syndrome measurement, the second and future syndrome measurements give the correct syndrome 1110.

Image: screenshot from Gottesman Ch. 12

Last time

We carried out a high-level analysis to illustrate the idea of a code's threshold error rate.

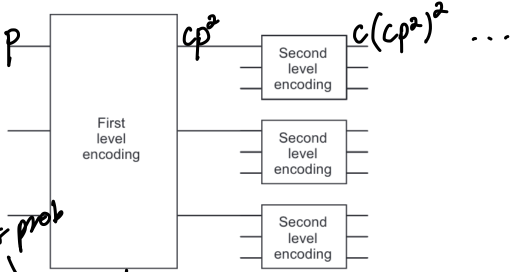
We estimated the probability of *two or more* errors in the encoded first qubit at the output.



Screenshot: Nielsen & Chuang, chapter 10.6.1 (Fig. 10.21)

Last time

Concatenation can suppress failure probability to be arbitrary low.



overall logical error prob

$$\frac{(cp)^{2^k}}{c} \leq \frac{\varepsilon}{\text{poly}(n)}$$

(target accuracy)

(number of gates in circuit)

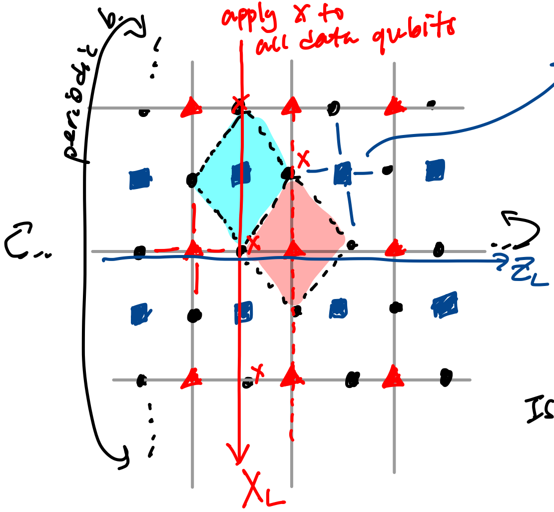
Can find k as long as $p < p_{th} = 1/c$ (*threshold condition*). The size of the simulation circuit becomes

$$O\left(\text{poly}\left(\log\left(\frac{\text{poly}(n)}{\varepsilon}\right)\right) \cdot \text{poly}(n)\right)$$

Learning outcomes

1. Define stabilizer generators for a family of 2D surface code and identify its logical Pauli operators
2. Construct circuits for surface code syndrome measurement
3. Carry out simulations in stim to estimate the surface code (pseudo)threshold

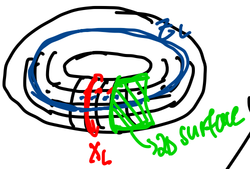
Preamble: the toric code



$$Z_{face} = \bigotimes_{f \in face} Z_f$$

$$X_{vertex} = \bigotimes_{v \in vertex} X_v$$

generators



Is it a stabilizer code?

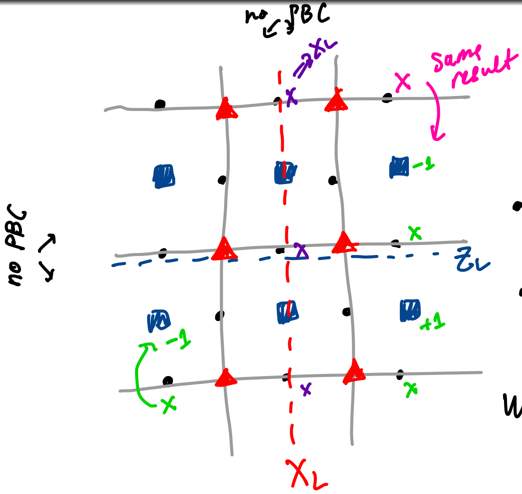
- they commute ✓

● data qubits

▲ X measure

■ Z measure

Unrotated surface code



X → + red
Z → □ blue

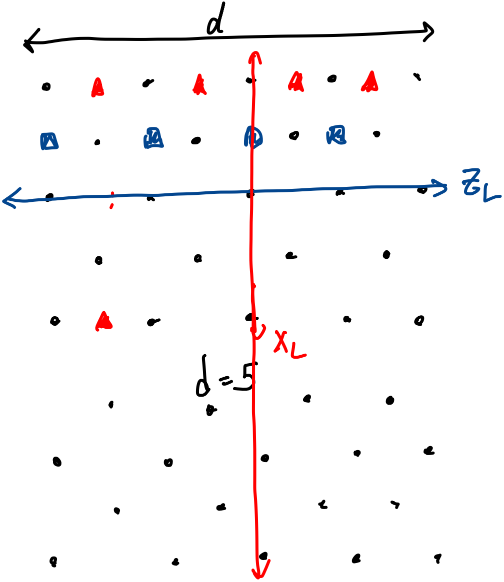
- how many data qubits?
  13
- how many stab. gens.?
  12

What is the distance?

- 1 is correctable
- 2 is detectable
- 3 is logical error

= d

Unrotated surface code: logical Pauli operators



$d = 5$

Data: $5^2 + 4^2 = 41$

Measure: 40

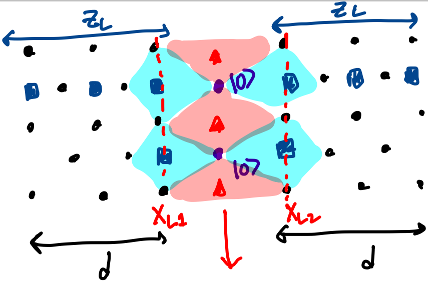
81 qubits

$d^2 + (d-1)^2$

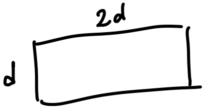
$+ d^2 + (d-1)^2 - 1$

$\Rightarrow O(2d^2)$

Unrotated surface code: merging



random outcome +1 / −1 of $X_{L1} \otimes X_{L2}$

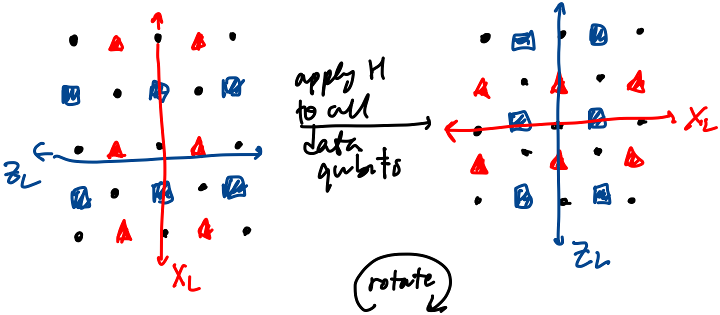


→ distance d

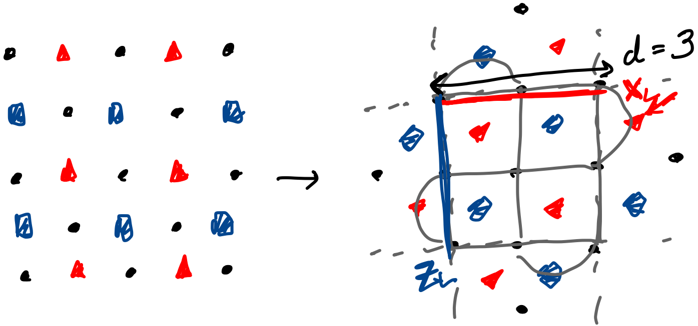
Unrotated surface code: splitting

☆ will add to
supplementary exercises

Unrotated surface code: Hadamard



Rotated surface code

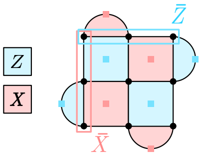


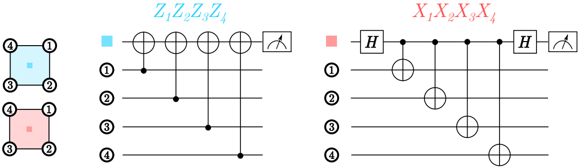
$\Rightarrow$ 13 data
12 measure

$\Rightarrow$ 9 data
8 measure

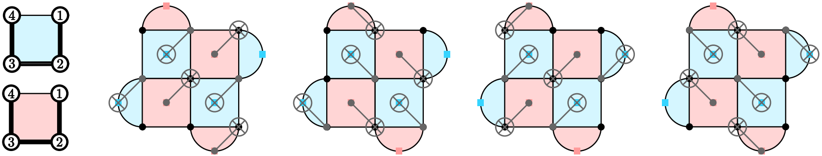
Syndrome measurement

We will start here Thursday

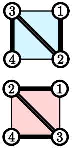


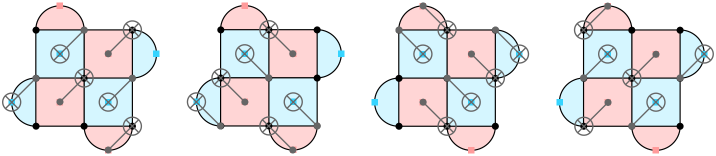


Syndrome measurement



Syndrome measurement





Stim simulations

Let's use stim to perform some experiments with various error models and estimate the threshold of the surface code.

Next time

Next class:

- The surface code part 2: decoding

Action items:

1. Work on project proposal
2. Watch PrairieLearn for exam information

Recommended reading for this / next class:

- Fowler, Mariatoni, Martini, and Cleland (2012), *Surface codes: Towards practical large-scale quantum computation*, Phys. Rev. A 86 (3) 032324.
- Horsman, Fowler, Devitt, and van Meter (2012), *Surface code quantum computing by lattice surgery*, NJP 14 123011.
- PennyLane tutorial on FT threshold
- surface code blog post